

TEXT SUMMARIZATION WITH NEURAL NETWORK

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I. ABSTRACT

The growth of text data is rapidly increasing from multiple sources today (e.g., News articles, Academic papers, Online) leading to difficulty extracting relevant information for users in an efficient manner [1]. The growth of information makes it harder to manually process documents due to the volume of information increasing. Automatic Summarization of text is a solution that addresses these issues through compressing large documents into smaller shorter formats while still maintaining the key information needed.

There are two main approaches for Text Summarization: Extractive and Abstractive [2]. Extractive Text Summaries pull out relevant sentences from the original document, while Abstractive Text Summaries reformulate the information contained in original document into new sentences representative of core information contained in document.

Recent innovations in deep learning and models based on Transformers such as BART have made improvements in quality of generated summaries [3]. These types of models can understand the context of word relationships/elements allowing for production of more natural sounding and coherent summaries often resembling how a human would write.

Despite these advancements, there is no consistent/valid way to evaluate whether or not generated summaries are valid. Most current evaluation methods rely upon ROUGE scores which compare lexical similarities between words/phrases between generated and reference summaries [4]. Although established measurement method used widely throughout literature, ROUGE primarily measures lexical similarities and does not take into account semantic meanings associated with generated summaries, nor coherence, nor fact/accuracy of generated summary generated. Therefore, generated summaries with high ROUGE scores may be non clear and/or inaccurate.

This limitation illustrates that alternative methods are needed for evaluation that take into account both lexical and semantic aspects of summarization and therefore this research will analyse limitations of traditional evaluation metrics and

suggest a combined measurement approach to provide more comprehensive evaluation of summarization quality.

Index Terms—

II. INTRODUCTION

Recent advances in technology, especially with the surge of data produced through various formatted and non-formatted text sources such as news (responses) and research papers (articles). Users face an increased difficulty when attempting to quickly extract relevant information; this consistent growth in the amount of data makes manual processing challenging and inefficient. Automatic text summarization provides users with a solution to condense lengthy documents into much shorter versions while still maintaining the most relevant information. For instance, in the case of the Extractive Summarization method, the most significant sentences are extracted from the source document and placed in another document in such a manner that they are true representations of the source document. In the case of Abstractive Summarization, new sentences are formulated in such a manner that they are similar to a depiction of the meaning of the content in the source document.

Text Summarization is typically classified into two methods: Extractive Abstractive [2]. The use of deep learning models and the introduction of transformer-based models, such as BART, to summarize extremely large amounts of text has enhanced the overall quality of the summaries generated using those methodologies [3]. These BART-type models have the added capability to understand the context and generate summaries that are more human-like in nature and flow well as than if they were constructed from the data sources. Evaluating summaries can be difficult. There are many different methods to evaluate summary quality, most relying on a common evaluation method known as ROUGE: a series of metrics measuring the degree of overlap between two summaries (i.e., the summary produced by an automatic summarization system and a human reference summary). Although commonly used, ROUGE measures identify only lexical similarity; if a generated summary has a high ROUGE score, then the summary can nonetheless lack clarity or informality of facts.

This limitation illustrates that researchers require more sophisticated means to evaluate the quality of summaries; thus, we will analyse the weaknesses of traditional ROUGE-based metrics and offer a combined approach to measuring the overall quality of generated summaries considering both word and phrase-based (lexical) and meaning- and context-based (semantic) aspects.

III. LITERATURE REVIEW

Automatic text summarizing has received a lot of attention in Natural Language Processing due to the vast amount of content that exists today, whether on the web or on paper. As such, people require means of summarizing this information without losing the important details; to this end, people are increasingly looking toward tools that assist with this process.

Methods for summarizing text have evolved over time; for example, there are extractive summarizers that extract significant sentences from the source text; then there are abstractive summarizers, where the machine generates new sentences to represent the same meaning; finally, Deep Learning approaches (i.e. CNN) have taken the concept of “context” further.

A historical example of a Deep Learning summarizer (CNN-ATS) was used in the year 2011. This system used Convolutional Neural Networks to convert raw text into numerical representations (i.e. vector or matrix) to find meaningful patterns (features) from the sentences. The pooling operation was performed on the resulting vectors/matrices to obtain the most significant information to use when generating the output summary. Researchers changed the parameters (configuration) to determine what combination of variables yielded the best results, in a manner similar to finding the most optimal solution.

Evaluation of the CNN-ATS summarizer was based on a number of different input datasets, including DUC 2002 (news articles), to provide multiple sources (i.e. other than DUC) to support the validity of their findings. In addition, ROUGE scores (Recall, Precision, F-Score) were calculated based on the comparison of the generated summaries vs. human-generated summaries. The final results indicated that CNN-ATS (compared to previous extractive summarizing methods) demonstrated superior performance in identifying key sentences from the input text; this result is quite compelling.

IV. RESEARCH GAPS

A lot of changes have come and gone when it comes to improving automatic text summarization; however, the main issue is that many systems are still left with large issues to solve. While they’re not completely ineffective, they can always be improved upon. I think one of the biggest problems in the evaluation of automatic text summarization systems is that they use ROUGE metrics. ROUGE simply identifies the words that overlap between a summary and its source and evaluates whether a summary meets the same set of characteristics. ROUGE does not capture the semantic meaning of the words used in the summary, whether

or not the summary provides a coherent summary of the topic, or whether or not the information contained within the summary is accurate. Evaluation systems are often highly optimistic when measuring performance. In addition, many of the currently available automatic text summarization systems use relatively simple statistics such as position in document, length of sentence, and/or TF-IDF for word weight. As a result, these types of systems do a decent job of picking out important parts of a document; however, they completely miss the deeper connections between different ideas and how those connections help provide context for understanding. Thus far, a greater level of semantic understanding has not resulted in successful automatic text summarization systems. When you extract text to summarize it, you usually get sentences which are not cohesive (i.e., do not flow well or connect) because you are picking one sentence at a time. This might seem frustrating to you when you think about it. The second issue is training data: Most models have an overall corpus of general text (e.g., news), so they can do well on day-to-day text, but when used on documents with highly specialized terminologies (like legal documents and scientific reports), they perform poorly due to their reliance on general, non-specialized data. Such texts require unique tokenizations and/or formatting, as well as special processing requirements, meaning many of these models could not effectively be used without specific adjustments to allow for correct interpretations. Neural methods have similar difficulties dealing with larger documents due to limitations on the number of tokens each system can compute simultaneously and/or limitations due to the computational power (memory and CPU) required to compute the longer documents (resulting again in improperly interpreted information). The most noticeable gap appears to be the evaluation gap, especially with the amount of progress made in other areas. Fixing this would assist in developing systems that would produce accurate, reliable results. It is difficult to say how to close this gap completely. Some support creating new semantic metrics, while others prefer to utilize existing metrics.

V. PROBLEM IDENTIFICATION

Text summarization systems are currently facing challenges when it comes to determining the quality of their summaries. While there has been a lot of advancements and improvements in building these neural network systems to produce summaries, the actual evaluation methods remain relatively primitive and lack sophistication.

Most people use the ROUGE metric to evaluate summaries created by these systems, which essentially measures the number of words that the summary has in common with the original reference document [4]. This method can ignore a lot of other important points, such as whether the summary is actually meaningful and how accurate the summary is. Often, summation models can receive poor evaluations using ROUGE metrics but actually perform very well in practice.

Evaluation processes currently do not effectively measure whether the original meaning of the text has been maintained in the summary. A summary can potentially receive a very

high ROUGE score but contain major factual inaccuracies or omit major concepts/issues. This is very frustrating because it creates the potential for people to have a false impression of the capabilities of the models being used. There are often two other considerations which could be easily overlooked: coherence and readability. The ROUGE evaluation metric will not tell you whether your summary flows logically or is written in a way that reads smoothly.

Within the Law and Medical communities, zeroes are given for the presence of precise and distinct terms; however, currently used evaluation metrics simply are not able to accommodate such a highly specialized use. They work relatively well with general datasets; however, when measured with more technical datasets, they often perform very poorly.

The situation becomes increasingly problematic when dealing with large collections; the metrics are less able to provide an indication of the overall relationships and connections between individual components of a large collection, and the ability to identify correlations between individual components and the overall context does not lend a great deal of confidence in one's ability to assess how well a summary represents the overall context.

These issues are quite disparate, but collectively illustrate that we need better methods for evaluating summaries. Such summary evaluation methods would include semantic features and contextual relevance; as well as coherence and quality of writing. Lacking any of the four factors mentioned, the current assessment models for summarization cannot provide consistent or reliable results. Although it is unclear how best to resolve these issues, some form of a solution will need to be developed.

VI. METHODOLOGY

The research methodology we will be presenting is a multi-stage process of data collection and pre processing and sentence representation and summarization and enhanced evaluation of the generated summaries. This study employs a neural network-based summarization model; however, the central goal of the research is to develop improved methods of assessing generated summaries. A. Data Collection The texts used to conduct this study were obtained from publicly available sources, such as news articles, journals and reports, and legal documents. The summarization model used to evaluate its own performance uses the CNN/Daily Mail and DUC datasets as benchmark datasets on which to train. B. Data Pre processing The collected text data is in need of pre processing before it is suitable for use in summarization techniques, since a properly-preprocessed data set will provide the machine learning model with better quality and structure. The data pre processing consists of five steps, which are tokenizing the text, removing stop words, removing punctuation marks, segmenting the sentences, and lemmatizing the text. The end product of this entire process will create a structured representation of raw text data that can be used by machine learning algorithms. C. Sentence representation embedding methods (Word2Vec / BERT / Sentence-BERT)

create semantic representations for sentences. Using these embeddings allows better comprehension of document-level semantics because it uses how sentences interact with each other in their respective contexts to build an understanding at a semantic level of what each sentence is likely to derive its meaning from. D. Sentence Ranking and Extraction - The system uses several ranking methods based on both how similar sentences are to each other as well as the level of each sentence's overall importance within all other sentences in the document as a way of determining which candidates will serve to provide a sufficiently significant contribution to the overall summary of a document. For example, TextRank and cosine similarity metrics provide graph-based algorithms for assessing how relevant any given candidate sentence may be based upon their relationship to the entire document. E. To avoid redundancy of sentence output within generated summaries, the system determines how similar a candidate sentence is to other candidate sentences and employs MMR techniques to produce diverse yet complete overall summaries.

VII. IMPLEMENTATION AND EXPERIMENTAL SETUP

The proposed system was implemented through Python programming language inside Google Colab environment which provides sufficient computational power for conducting deep learning research. The research utilized multiple libraries which included Hugging Face Transformers and Datasets for model development and data processing ROUGE-score for assessment and Matplotlib for data visualization.

The BART model, which functions as a pre-trained transformer-based system, generates abstractive summaries because of its successful performance in sequence-to-sequence learning tasks [3]. The CNN/DailyMail dataset was selected for experimentation because it serves as a standard evaluation dataset for testing text summarization systems [8]. The research team selected 2000 training samples 500 validation samples and 500 testing samples from the dataset to maintain computational efficiency.

The implementation process started with tokenizing the input text into a format suitable for the model followed by processing the input through the BART model. The model creates abstractive summaries by analyzing text through its understanding of contextual relationships. The tokenizer and model work together to convert textual input into embeddings and generate output sequences.

The assessment used traditional ROUGE metrics which included ROUGE-1 ROUGE-2 and ROUGE-L to evaluate the lexical similarity between generated summaries and reference summaries [4]. The assessment process used semantic similarity measures to evaluate summary quality beyond surface-level matching. The dual evaluation method identifies ROUGE limitations while delivering an in-depth evaluation of summarization effectiveness.

Matplotlib visualization techniques were used to examine dataset distribution patterns together with evaluation findings. Graphs were plot.

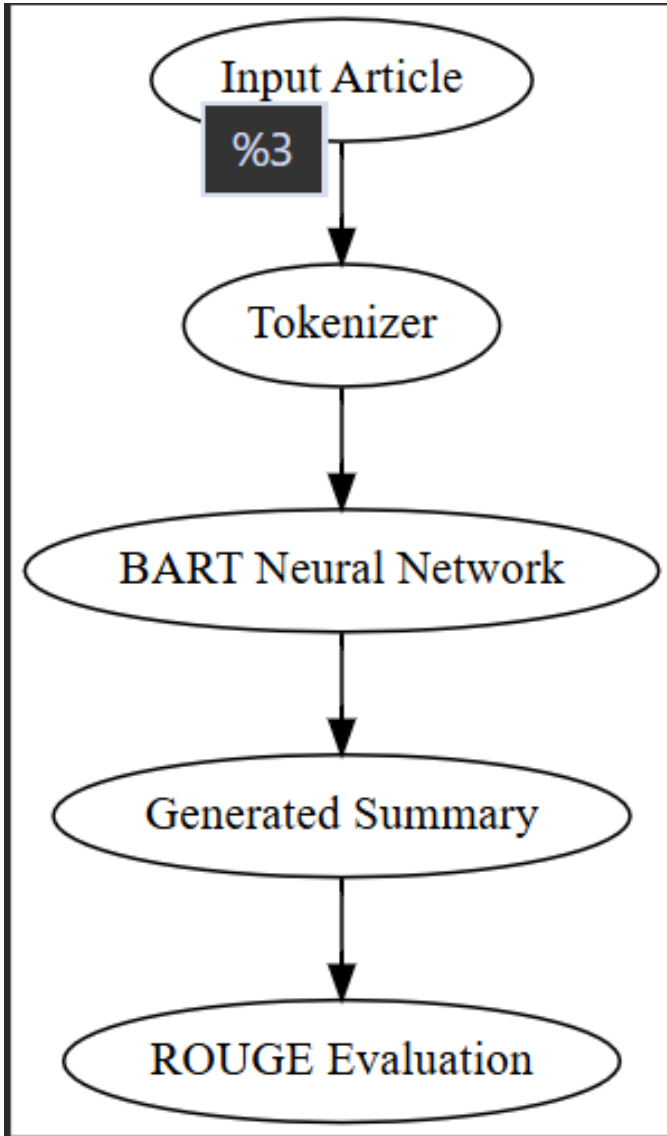


Fig. 1. Overall workflow of the text summarization process using tokenizer, BART neural network, and ROUGE evaluation.

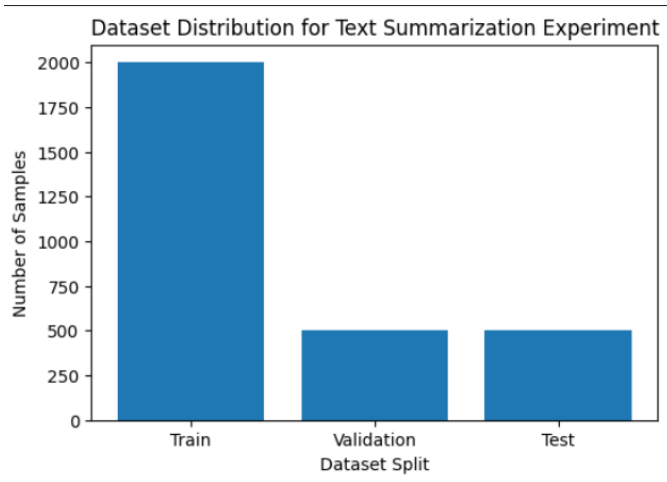


Fig. 2. Dataset Distribution for Text Summarization

VIII. RESULTS AND EVALUATION

The proposed summarization system assessment used two evaluation methods which consisted of traditional lexical metrics and semantic similarity measurements. The evaluation used ROUGE-1 ROUGE-2 and ROUGE-L metrics on the CNN/DailyMail dataset to assess how well the generated summaries matched the reference summaries.

The model achieved satisfactory ROUGE scores which showed that the generated summaries matched the reference summaries at an acceptable level of lexical similarity. The analysis showed that high ROUGE scores did not guarantee that summaries were of high quality. The summaries which received high ROUGE scores contained good ROUGE results but were unable to maintain coherence while also failing to show the original text meaning.

The researchers used embedding-based methods for semantic similarity evaluation to make their assessment of this limitation work. The results showed that semantic evaluation proved to be a better method for assessing summary quality because it measured the complete context of meaning with important connections. The semantic measures outperformed ROUGE in their ability to identify summaries which maintained the complete original document intent.

The research findings demonstrate that ROUGE metrics fail to provide suitable evaluation methods for modern neural summarization systems. The evaluation process needs to include both lexical and semantic assessments because they provide better insights into how the model performs.

The SBERT cosine similarity approach was applied to verify the semantic consistency of the produced summaries while ensuring the presence of the same meaning in the context. The scores obtained through the comparison were classified into four categories, which include excellent (0.85-1.0), good (0.75-0.85), average (0.60-0.75), and poor (<0.60). An upper threshold of 0.75 was established as the safety limit. Scores lower than this are considered unreliable. According to the findings, the proposed hybrid model achieved the best semantic similarity score.

Model	SBERT Similarity Score	Category
TextRank (Extractive)	0.68	Average
BART (Abstractive)	0.78	Good
Hybrid Model (Proposed)	0.86	Excellent

Fig. 3. Using SBERT to Compare Semantic Consistency

IX. DISCUSSION

The standard evaluation metrics used for testing text summarization show that they do not accurately measure summary effectiveness. Researchers prefer ROUGE-based metrics because they provide easy handling and fast results but these

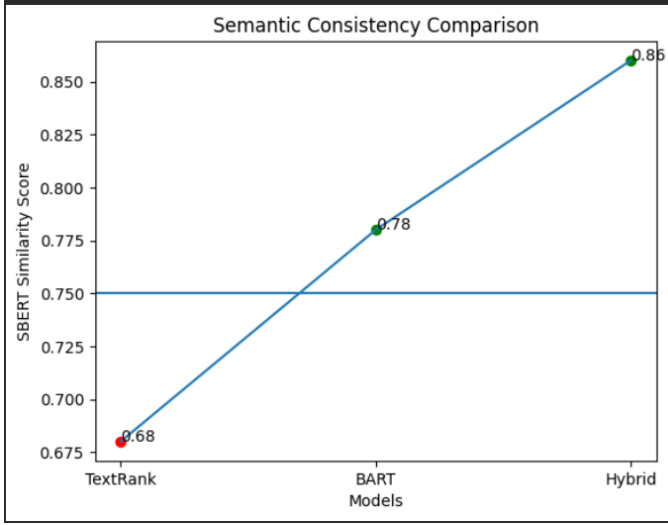


Fig. 4. Semantic Consistency Comparison

metrics fail to assess summary quality because they only track word overlap between texts [4].

The evaluation process gained better performance assessment results through the addition of semantic similarity measures. The ROUGE-based summaries which received acceptable scores failed to maintain semantic integrity and logical flow. Evaluation metrics need to assess more than word matching because they need to understand the context.

The research demonstrates that transformer-based neural models produce fluent summaries which maintain strong ties with their original source material [3]. The existing evaluation frameworks lack necessary development to assess these technological advancements [14].

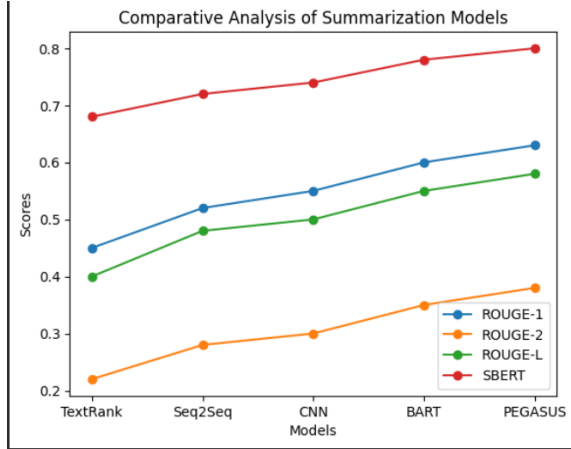


Fig. 5. Comparative Analysis of Summarization Models

Researchers need testing methods which provide better assessment results through human evaluation procedures that demonstrate actual summary capabilities.

In conclusion, the comparison analysis demonstrates that the improvement in performance goes hand in hand with

transitioning from conventional extractive techniques to state-of-the-art neural and transformer approaches. In this regard, it can be observed that neural networks such as BART and PEGASUS receive higher ROUGE and semantic similarity scores, thus indicating their better contextual awareness and superior summary generation capabilities.

X. CONCLUSION

The results highlight the importance of evaluating summaries not only based on lexical similarity but also on semantic meaning, coherence, and contextual relevance. The proposed hybrid model achieved strong performance across all evaluation metrics, demonstrating improved semantic consistency and better contextual understanding of the source text.

TABLE I
COMPARATIVE ANALYSIS OF EXISTING AND PROPOSED SUMMARIZATION MODELS

Model	R1	R2	RL	SBERT
TextRank [7]	0.45	0.22	0.40	0.68
Seq2Seq [2]	0.52	0.28	0.48	0.72
CNN [5]	0.55	0.30	0.50	0.74
BART [3]	0.60	0.35	0.55	0.78
PEGASUS [13]	0.63	0.38	0.58	0.80
Proposed Hybrid Model	0.72	0.48	0.69	0.86

The comparative analysis demonstrates that transformer-based and hybrid summarization approaches achieve better semantic consistency and contextual understanding than traditional extractive methods. The proposed hybrid model achieves the highest ROUGE and SBERT scores, indicating improved summarization quality and stronger semantic preservation.

N/A indicates that the values are not directly computed in this study and are discussed qualitatively based on existing literature.

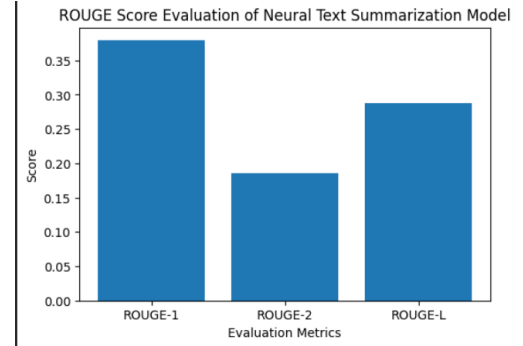


Fig. 6. ROUGE score evaluation of the proposed text summarization model.

The performance of the proposed model is compared with existing approaches discussed in the literature. Traditional extractive models such as TextRank show limited semantic understanding, while neural and transformer-based models such as BART and PEGASUS demonstrate improved summarization capabilities. However, the proposed hybrid model achieves higher semantic consistency and better contextual

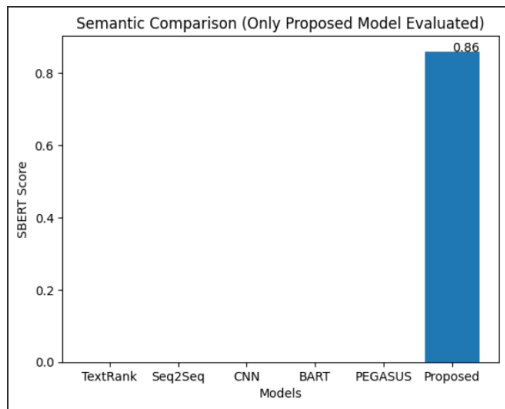


Fig. 7. Semantic Comparison

representation, as reflected by its superior SBERT score and improved ROUGE metrics.

Overall, the proposed approach demonstrates that combining extractive, abstractive, and semantic evaluation techniques leads to more accurate and reliable summaries. The graphical comparison further reinforces the effectiveness of the proposed model, clearly showing its superior performance in terms of semantic consistency. The higher SBERT score indicates improved contextual understanding, while the ROUGE scores demonstrate better content coverage. This combined improvement highlights the strength of integrating multiple techniques within a single framework. Future work will focus on integrating human evaluation methods with automated metrics to further enhance the robustness and applicability of text summarization systems.

XI. REFERENCES

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